

Pinned group summary: $SU(n=2, q=1, \epsilon=1)$

Pinning-Sympy automated output

July 6, 2026

Contents

1	Basic attributes	2
1.1	Hermitian form	2
1.2	Generic Lie algebra element	2
1.3	Generic torus element	2
1.4	Trivial characters	2
2	Root system	3
2.1	Dynkin diagram	3
2.2	Cartan matrix	3
2.3	Roots and coroots	3

1 Basic attributes

Group	SU(n=2, q=1, eps=1)
Matrix size	2×2
Rank	1

1.1 Hermitian form

Dimension	2
Witt index	1
Primitive element	$\sqrt{d}$
Epsilon	1

Matrix: $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{r00} & \sqrt{d}x_{i01} \\ \sqrt{d}x_{i10} & -x_{r00} \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 \\ 0 & \frac{1}{t_0} \end{bmatrix}$$

1.4 Trivial characters

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	A1
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram

$$\textcircled{0}$$

2.2 Cartan matrix

$$[2]$$

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
$(-2, 0)$	$(-1, 1)$	No	$-$	No
$(2, 0)$	$(1, -1)$	Yes	$+$	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

No pairs of roots generate positive integer linear combinations.